

OPTICROP: SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE

TESTING AND VERIFICATION REPORT

1. Model Accuracy Verification

To verify system reliability, we compared four classification models on the crop dataset:

Algorithm	Training Accuracy	Test Accuracy	Weighted F1-Score
Random Forest Classifier	100.0%	100.0%	1.00
Decision Tree Classifier	100.0%	98.1%	0.98
K-Nearest Neighbors (KNN)	98.5%	97.3%	0.97
Logistic Regression	96.2%	95.1%	0.95

2. Web Interface and System Checks

- Forms: Verified that submitting empty fields triggers error responses.
- Latency: Tested local server predictions, returning HTML templates in under 50ms.
- Data Persistence: Verified that predictions are successfully written to the CSV logs.